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Dear Hiring Manager,

As a final-year Mechanical Engineering student at the University of Sydney with a strong parallel interest in IT systems, I offer a combination of hands-on engineering experience and practical expertise in homelab management, UNIX environments, and networking.

My homelab, built from repurposed hardware including an Intel i5-11400F, GTX 1060, and 32 GB of RAM from decommissioned servers, serves as both a NAS and a media server running TrueNAS for reliable storage and network access. Previous iterations included Windows Hyper-V hosting Ubuntu Server with Docker containers for cloud and media services, and OpenWrt for networking—later migrated to a Raspberry Pi 4. This process gave me extensive exposure to virtualization, containerization, and network configuration, while also teaching valuable lessons in stability, system recovery, and service uptime.

In addition, I have two years of experience daily-driving Linux, dual-booting distributions including Ubuntu, Debian, and Arch. During this time, I explored system customization, performance optimization, and idle load minimization—skills directly transferable to optimizing IT infrastructure and streamlining resource usage. My comfort in both UNIX and Windows environments ensures adaptability across varied system landscapes.

Alongside my IT capabilities, my engineering background has equipped me with problem-solving discipline and precision. In roles such as Lead Aerodynamics Engineer for Sydney Motorsports and CFD Engineer at Dovetail Electric Aviation, I have used high-performance computing tools, scripting (Python, MATLAB), and optimization workflows—experiences that parallel IT infrastructure management in their need for efficiency, accuracy, and scalability.

Key strengths include:

- **Systems & Networking:** Homelab architecture, NAS management, containerized services (Docker), OpenWrt network configuration.
- **UNIX/Linux Expertise:** Daily use of Ubuntu, Debian, and Arch; performance tuning; system recovery and customization.
- **Engineering & IT Crossover:** Automation, computational optimization, scripting, and integration of hardware/software workflows.

I am eager to contribute these skills to your company, supporting the design, maintenance, and optimization of robust IT systems. I welcome the opportunity to discuss how my blend of IT and engineering expertise can add value to your team.

Thank you for considering my application. I look forward to the possibility of joining your organisation.

Sincerely,
Winston Zhang